

Data Analysis and Visualization

PART III: Analyzing data, Interpreting the analyses, Communicating the results statistics

Module: Inferences about population proportion and central values

Hypothesis Testing

Lecturer: Diu Tran, PhD

Department of Computer Science
Faculty of Information Technology
Ton Duc Thang University

Ho Chi Minh, Week 7, 2025

Content

- 1 Statistics Fundamentals
- 2 Hypothesis Testing for Population Proportions
- 3 Hypothesis Testing for Population Mean
- 4 Hypothesis Testing for the Difference Between Two Population Proportions
- 5 Hypothesis Testing for the Difference Between Two Population Means
- 6 Lab 7

Statistics Fundamentals

For your reference: <https://www.youtube.com/statquest>

- Video: Hypothesis Testing and The Null Hypothesis, Clearly Explained!!!
- Video: Alternative Hypotheses: Main Ideas!!!
- Video: p-values: What they are and how to interpret them
- Video: How to calculate p-values
- Video: p-hacking: What it is and how to avoid it!

Hypothesis testing for population proportions

- Example research of interest:
 - ▶ In previous years, 52% of parents believed that electronic devices and social media were the cause of sleep disorders in young people (aged 10–19).
 - ▶ Nowadays, whether there are more parents believe that these electronic devices and social media cause sleep disorders in young people?
- Analysis:
 - ▶ Population: parents with children aged 10–19 years
 - ▶ Parameter of interest: The proportion p of parents who believe that electronic devices and social media cause sleep disorders in their children.
- Goal: We aim to test whether there has been an increase in the percentage of parents who believe electronic devices and social media cause sleep disorders in their children.
- Note: It is crucial to define the hypothesis clearly before collecting data to ensure an unbiased test.

Step 1: Formulating the Statistical Hypotheses

- **Parameter of Interest (p):**

- ▶ p is the **population proportion** of all parents with teenage children who believe that electronic devices and social media cause a lack of sleep.

- **Hypotheses:**

- ▶ **Null Hypothesis (H_0):** The proportion is equal to the established value.

$$H_0 : p = 0.52$$

- ▶ **Alternative Hypothesis (H_1):** We want to investigate if the proportion has significantly increased.

$$H_1 : p > 0.52$$

- **Significance Level (α):**

- ▶ This is the probability of rejecting H_0 when it is true.

$$\alpha = 0.05$$

Step 2: Data Check & Sample Statistic

- **Sample Information:**

- ▶ A **random sample** of $n = 1018$ parents with teenage children was surveyed.
- ▶ The **sample proportion** ($\hat{p}$) who believe electronic devices and social media cause lack of sleep is:

$$\hat{p} = 0.56$$

- **Data Collection Requirements:**

- ▶ **Random Sampling:** Ensure the sample was collected using a ****random method****.
- ▶ **Sample Size Condition (Normality):** Ensure the sample size is large enough so that the sampling distribution of $\hat{p}$ is approximately normal.
 - ★ This requires at least 10 successes and 10 failures in the sample, using the hypothesized proportion p_0 :

$$np_0 \geq 10 \quad \text{and} \quad n(1 - p_0) \geq 10$$

Step 2: Calculating the Test Statistic

- **General Formula for the Test Statistic:**

$$\text{Test Statistic} = \frac{\text{Best Estimate} - \text{Hypothesized Value}}{\text{Standard Error}}$$

- Values used: Best estimate (Sample Proportion) $\hat{p} = 0.56$;
Hypothesized Value (Null Hypothesis) $p_0 = 0.52$
- Standard Error Assumption: Under the assumption that **H₀** is true, the ****Standard Error**** of $\hat{p}$ is:

$$\text{SE}(\hat{p}) = \sqrt{\frac{p_0(1 - p_0)}{n}}$$

- Z-Test Statistic Calculation (for Proportions): $\mathbf{Z} = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$

$$Z = \frac{0.56 - 0.52}{\sqrt{\frac{0.52(1 - 0.52)}{1018}}} = \frac{0.04}{\sqrt{\frac{0.2496}{1018}}} \approx \frac{0.04}{0.0157} \approx 2.555$$

Interpreting the Z-Test Statistic

- Theoretical Basis (Central Limit Theorem):
 - ▶ The sampling distribution of $\hat{p}$ is approximately normal:

$$\hat{p} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- ▶ Under H_0 (i.e., $p = p_0$), we have:

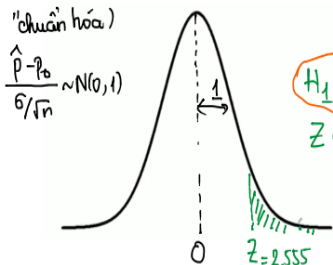
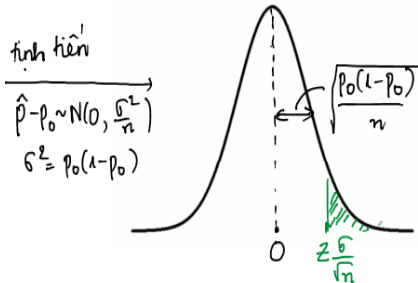
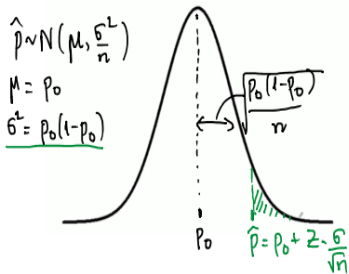
$$\mu = p_0 \quad \text{and} \quad \sigma^2 = p_0(1 - p_0)$$

- ▶ Therefore, the standardized test statistic is approximately Standard Normal:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N(0, 1)$$

- The value **$Z = 2.555$** means the observed sample proportion ($\hat{p} = 0.56$) is **2.555 standard errors** above the hypothesized population proportion ($p_0 = 0.52$), assuming the Null Hypothesis (H_0) is true.

⊗ Bước 3 : tính p-value từ giá trị thống kê kiểm định (test statistic)



$H_1 : p > 0.52$

$z \sim N(0, 1) \rightarrow \text{trabảng } P(z > 2.555) = 0.0053$

p-value = 0.0053

Step 3: Finding the p-value

- **Alternative Hypothesis:** The test is a ****one-tailed (right-tailed) test**** because we are looking for an increase:

$$H_1 : p > 0.52$$

- **Test Statistic Result:** $Z = 2.555$
- **Definition of p-value:** The p-value is the probability of observing a test statistic (Z) at least as extreme as the one calculated, assuming the null hypothesis (H_0) is true.
- **Calculation using the Standard Normal Distribution:** For a right-tailed test, the p-value is the area to the right of the calculated Z-score:

$$\text{p-value} = P(Z \geq 2.555)$$

- **Result (from Z-table or statistical software):**

$$\text{p-value} = 0.0053$$

Step 4: Decision & Conclusion

- **Recall our values:**

- ▶ **p-value** calculated in the previous step: **0.0053**
- ▶ **Significance Level (α)** set in Step 1: **0.05**

- **Decision Rule:**

- ▶ If **p-value** $\leq \alpha$, we **reject** the Null Hypothesis (**H_0**).
- ▶ If **p-value** $> \alpha$, we **fail to reject** the Null Hypothesis (**H_0**).

- **Comparison:** $0.0053 < 0.05$ i.e., $\text{p-value} < \alpha$

- **Statistical Decision:** Based on the comparison, we **reject the null hypothesis ($H_0 : p = 0.52$)**.

- **Conclusion:** There is sufficient evidence to conclude that the population proportion of parents with teenage children who believe electronic devices and social media cause their children to lack sleep is ****greater than 52%****.

SUMMARY: The 4 Steps of Hypothesis Testing

① State Hypotheses & Significance Level:

$$H_0 : p = 0.52 \quad H_1 : p > 0.52 \quad \alpha = 0.05$$

② Check Data & Calculate Test Statistic: For a population proportion, the Z-test statistic is:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = 2.555$$

③ Convert Test Statistic to p-value: $= P(Z \geq 2.555) = 0.0053$

④ Compare & Conclude:

- ▶ Comparison: p-value (**0.0053**) < α (**0.05**)
- ▶ Decision: Reject H_0
- ▶ Final Conclusion: There is ****sufficient statistical evidence**** to support the alternative hypothesis (**H_1**), meaning the proportion of parents who believe electronic devices and social media cause their children to lack sleep is ****greater than 52%****.

Hypothesis Testing for Population Mean

Example Situation:

From past experience and previous observations (e.g., we once observed that the average jumping distance of adults is about 82 inches), we are interested in the following hypothesis question:

Hypothesis Question

"Is the average jumping distance of adults greater than 80 inches?"

This is a **hypothesis** that we do not know whether it is true or not.

We look for **evidence or reasoning** to either:

- Reject this hypothesis, or
- Fail to reject it (not enough evidence to reject).

Problem Analysis

- **Population:** all adults.
- **Parameter of interest:** the population mean of the jumping distance of all adults (denoted by μ).

We conduct hypothesis testing based on a sample containing the jumping distances of 25 adults.

The hypothesis is proposed based on prior experience and the research objective.

A **statistical hypothesis** is formulated before collecting data, in order to avoid subjective bias when forming conclusions.

Step 1: Define the Hypotheses

Null and Alternative Hypotheses

Null hypothesis (H_0): The population mean jumping distance of adults (μ) is 80 inches.

Alternative hypothesis (H_1): The population mean jumping distance is **greater than 80 inches**.

Notation:

$$H_0 : \mu = 80$$

$$H_1 : \mu > 80$$

μ represents the population mean of adults' jumping distances.

Question: How much evidence is needed to reject H_0 and support H_1 ?

Significance level: $\alpha = 0.05$

Step 2: Collect, Process, and Examine the Data

Goal: Verify the sample assumptions and compute the **test statistic**.

- **Sample size (n):** 25
- **Minimum distance:** 63 inches
- **Maximum distance:** 115 inches
- **Mean distance:** 82.48 inches
- **Standard deviation:** 15.06

Assumptions to check:

- The sample is randomly selected and representative of the population.
- The distribution of jumping distances for adults is approximately normal. (For $n = 25$, the Central Limit Theorem suggests the sampling distribution of the mean is approximately normal even if the population distribution is not perfectly normal. Actually, using t -distribution is better.)

Interpreting the Sample Mean

- Observed sample mean: $\bar{x} = 82.48$ inches. Is it really “significantly larger” than the hypothesized population mean of 80 inches?
- A difference of 2.48 inches, is it large or small? If we repeated this experiment many times, we would get different sample means even if H_0 were true and the true population mean were 80 inches. Sample means fluctuate around the true population mean sometimes higher/lower.
- So, is 82.48 inches a meaningful difference or just a result of random variation? To measure how far the sample mean is from the hypothesized mean, we compute the **test statistic**, using the concept of the **standard error** $= SE = \frac{\sigma}{\sqrt{n}}$
- But since the population standard deviation σ is usually unknown, we estimate it using the sample standard deviation s :
Estimated standard error $= \widehat{SE} = \frac{s}{\sqrt{n}}$

Step 2: Compute the Test Statistic

Purpose: To measure how far the sample mean is from the hypothesized population mean under H_0 .

Assuming the sampling distribution of the mean is approx. normal:

$$t = \frac{\text{Best estimate} - \text{Value under } H_0}{\text{Estimated standard error}}$$

$$t = \frac{\bar{X} - 80}{S/\sqrt{n}} = \frac{82.48 - 80}{15.06/\sqrt{25}} = 0.82$$

Interpretation: The test statistic $t = 0.82$ indicates that the observed sample mean is only about **0.82 standard errors above** the hypothesized mean of 80 inches.

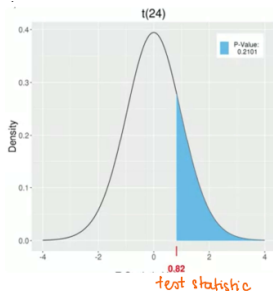
We use the **p-value** to determine how unusual (or rare) this value of t would be if H_0 were true.

Step 3: Determine the p-value

Purpose: Find the p-value to answer the question: *“If the null hypothesis H_0 is true, is the observed test statistic $t = 0.82$ unusual enough to reject H_0 ?”*

If H_0 is true: The test statistic t follows a t -distribution with degrees of freedom $df = n - 1 = 25 - 1 = 24$.

$$\text{p-value} = P(\text{blue area}) = P(t > 0.82) = 0.21$$



Understanding the p-value and Test Statistic

Recall the meaning of the p-value: The p-value represents the probability of observing a test statistic as extreme as (or more extreme than) the calculated value $t = 0.82$, assuming the null hypothesis H_0 is true.

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(24) \quad \Rightarrow \quad \bar{X} = \mu + 0.82 \frac{S}{\sqrt{n}}$$

Interpretation: If H_0 is true, a test statistic of $t = 0.82$ corresponds to a sample mean $\bar{X}$ that is **0.82 standard errors above the true mean μ** . We then ask: Is such a deviation likely (common) or unlikely (rare) under H_0 ?

Alternative Hypothesis Considerations:

- If $H_1 : \mu > 80$, focus on the right-hand tail (one-sided test).
- If $H_1 : \mu < 80$, focus on the left-hand tail.
- If $H_1 : \mu \neq 80$, consider both tails (two-sided test).

Step 4: Making the Decision About H_0

- **Decision rule:**

- ▶ **Reject $H_0 \Rightarrow$** The alternative hypothesis H_1 is supported.
- ▶ **Fail to reject $H_0 \Rightarrow$** There is not enough evidence to reject H_0 (but this does not mean H_0 is true).

- Given that:

$$p\text{-value} = 0.21 > \alpha = 0.05$$

- Therefore, we **fail to reject H_0** .

Interpretation:

- The test statistic $t = 0.82$ is not in the most extreme 5% of possible values (under H_0).
- In other words, the difference between the sample mean (82.48 inches) and the hypothesized population mean (80 inches) is not large enough to be considered statistically significant.
- Hence, based on our data, we do not have sufficient evidence to conclude that the average height of the population is greater than 80 inches.

Method 2: Using CI to Test the Hypothesis

- **Significance level:** $\alpha = 0.05 = 5\%$
- This corresponds to a **90% confidence interval** (two-tailed test).

Given data: $\bar{X} = 82.48$, $s = 15.08$, $n = 25$ $t^* = 1.711$

Confidence Interval:

$$\bar{X} \pm t^* \frac{s}{\sqrt{n}} = 82.48 \pm 1.711 \times \frac{15.08}{\sqrt{25}} = 82.48 \pm 5.15$$
$$\Rightarrow (77.33, 87.63) \text{ inches}$$

Interpretation:

- The 90% confidence interval for the population mean height is (77.33, 87.63) inches.
- Since 80 inches lies within this range, we **fail to reject** H_0 . Therefore, there is not enough evidence to support the alternative hypothesis H_1 (that the true mean height is greater than 80 inches).

Comments on the Normality Assumption

- The distribution of the sample data (distances) appears to be **left-skewed** and the sample size ($n = 25$) is relatively small (< 30), which makes the assumption of normality less reliable.
- If the normality assumption is questionable or not optimal, we can use a **non-parametric test**, which does not require normally distributed data.
- In such cases, an example of a suitable non-parametric test is the **Wilcoxon Signed-Rank Test**.

Summary

- Hypothesis testing is used to test a statement (hypothesis) about a population parameter of interest (e.g., population proportion, population mean).

Steps in Hypothesis Testing:

- 1 **State the hypotheses:** Formulate the null hypothesis (H_0) and the alternative hypothesis (H_1), and determine the significance level (α).
- 2 **Check assumptions and Compute the test statistic:** Verify sampling method (random sample, normality assumption), collect data, and compute descriptive statistics. Compute the test statistic (e.g., t -statistic)
- 3 **Compute p-value:** convert test statistic to p-value.
- 4 **Make a decision:** Compare the p-value with the significance level to decide whether to reject or fail to reject H_0 .

Hypothesis Testing for the Difference Between Two Population Proportions

Example Problem:

Is there a significant difference between the proportion of urban parents with children aged 6–18 who attend swimming lessons and the proportion of rural parents with children aged 6–18 who attend swimming lessons?

Analysis:

- Population 1: Urban parents with children aged 6–18.
- Population 2: Rural parents with children aged 6–18.
- Parameter of interest: $p_1 - p_2$, where p_1 is the proportion for population 1 and p_2 is the proportion for population 2.

Objective: To test whether there is a **statistically significant difference** between the two proportions (urban vs. rural parents) at a **10% significance level** ($\alpha = 0.10$).

Step 1: Formulate the Statistical Hypotheses

Null Hypothesis (H_0):

$$H_0 : p_1 - p_2 = 0$$

Alternative Hypothesis (H_1):

$$H_1 : p_1 - p_2 \neq 0$$

We want to determine whether the proportions are **significantly different**, regardless of whether $p_1 > p_2$ or $p_2 > p_1$.

Where: p_1 and p_2 are the population proportions of the two groups.

Significance Level:

$$\alpha = 10\% = 0.10$$

Step 2: Check Data and Calculate Test Statistic

Survey Data:

- 308 urban parents with children aged 6–18 were surveyed, of which 120 have children attending swimming classes.
- 247 rural parents with children aged 6–18 were surveyed, of which 91 have children attending swimming classes.

Checking the Data:

- The two samples are independent and randomly selected.
- Each sample size is large enough for the Central Limit Theorem to apply (at least 10 “yes” and 10 “no” responses in each sample).

Pooled Proportion: $\hat{p} = \frac{120+91}{308+247} = 0.38$

Sample 1: $308 \times 0.38 = 117$, $308 \times (1 - 0.38) = 191 (> 10)$

Sample 2: $247 \times 0.38 = 94$, $247 \times (1 - 0.38) = 153 (> 10)$

Step 2: Calculate the Test Statistic (Z)

Formula:

$$Z = \frac{\text{Best Estimate} - \text{Hypothesized Value}}{\text{Estimated Standard Error}} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)_{H_0}}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Best Estimates:

$$\hat{p}_1 = \frac{120}{308} = 0.39, \quad \hat{p}_2 = \frac{91}{247} = 0.37, \quad \hat{p}_1 - \hat{p}_2 = 0.02$$

Pooled Proportion: $\hat{p} = \frac{120+91}{308+247} = 0.38$

Standard Error:

$$SE = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = 0.041$$

Test Statistic:

$$Z = \frac{0.02 - 0}{0.041} = 0.48$$

Understanding Standard Error and Z-Test Statistic

We can use the “unpooled” standard error:

$$SE = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \approx 0.041$$

Interpreting the Z-test statistic:

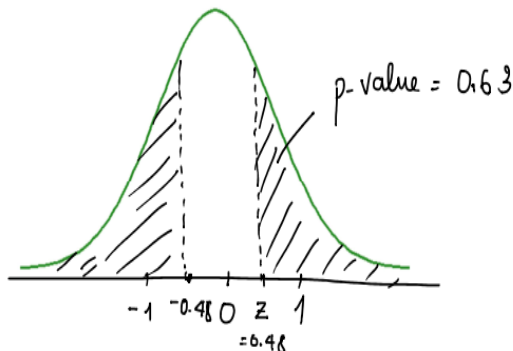
$$\text{Estimated difference} = \text{Hypothesized difference} + 0.48 \times SE$$

Our observed difference between the two sample proportions differs from the hypothesized difference (which is 0) by about 0.48 standard errors - meaning the observed difference is small and within the range expected under the null hypothesis.

Step 3: Compute the p-value from Z-test

$$Z \sim N(0, 1)$$

$$p\text{-value} = 2 \int_{-\infty}^{-0.48} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 2 \times 0.3156 = 0.63$$



Step 4: Compare p-value and significance level α and make a conclusion

$$p\text{-value} = 0.63 > \alpha = 0.10 \Rightarrow \text{Fail to reject } H_0$$

Conclusion: Based on the observed sample data and hypothesis testing, there is not enough evidence to reject the null hypothesis H_0 . Thus, there is **no significant difference** between the proportions of urban and rural parents with children aged 6–18 who attend swimming lessons.

Hypothesis Testing for the Difference Between Two Population Means

Research Question: Examine whether the average Body Mass Index (BMI) of adult males (aged 18–29) in Vietnam is significantly different from that of adult females.

Population: All adults aged 18–29 currently living in Vietnam.

Parameter of Interest:

$$\mu_1 - \mu_2$$

where μ_1 = mean BMI (kg/m^2) of adult males, μ_2 = mean BMI (kg/m^2) of adult females.

Step 1: State the null hypothesis and the alternative hypothesis; determine the significance level α

Step 1: Define Hypotheses and Significance Level

$$H_0 : \mu_1 = \mu_2 \quad (\mu_1 - \mu_2 = 0)$$

$$H_1 : \mu_1 \neq \mu_2 \quad (\mu_1 - \mu_2 \neq 0)$$

Significance level: $\alpha = 5\% = 0.05$

Step 2: Check the data; verify assumptions and compute the test statistic (e.g., t-test or z-test)

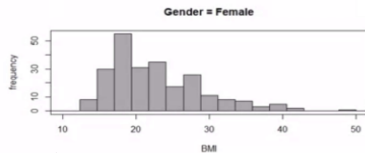
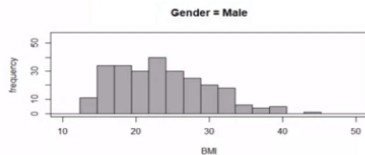
	Male	Female
Mean	23.57	22.83
St. Dev.	6.24	6.43
Min	12.5	12.4
Max	44.9	48.8
n	258	239

Check data assumptions:

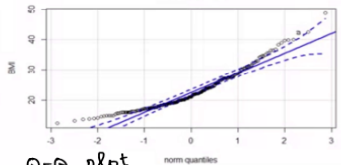
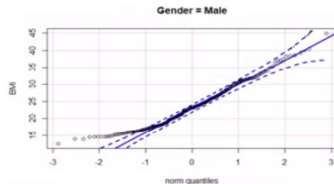
- Is the data collected randomly (random sample)? The samples should represent the population, and each person should have an equal probability of being selected.
- Are the two samples collected independently?

Checking Data Assumptions

-
- Do the observations $X_1, X_2, \dots, X_n$ (corresponding to random variables) follow a normal distribution? Either both populations are normally distributed, or the sample sizes are large enough for the sample means to be approximately normal.



histogram



Q-Q plot

Step 2: Calculate the Test Statistic

Hypotheses:

$$H_0 : \mu_1 - \mu_2 = 0 \quad H_1 : \mu_1 - \mu_2 \neq 0$$

Best estimate of $\mu_1 - \mu_2$:

$$\hat{\mu}_1 - \hat{\mu}_2 = 23.57 - 22.83 = 0.74 \text{ (kg/m}^2\text{)}$$

Interpretation: The observed difference between the two sample means is 0.74. We test whether this difference is significantly different from 0. The test statistic measures how far the sample statistic is from the hypothesized population parameter, scaled by its estimated standard error. The further this value is from 0, the more evidence we have against the null hypothesis.

General formula:

$$\text{Test statistic} = \frac{\text{Best estimate} - \text{Hypothesized value under } H_0}{\text{Estimated standard error}}$$

Step 2: Compute the Test Statistic

Case 1: Equal population variances (special case)

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$Z = \frac{(23.57 - 22.83) - 0}{\sqrt{\frac{257 \times 6.24^2 + 238 \times 6.43^2}{258+239-2} \left(\frac{1}{258} + \frac{1}{239} \right)}} = 1.30$$

Interpretation: The observed difference between the two sample means is 1.30 times larger than the hypothesized difference (0 kg/m²), measured in units of the standard error.

Case 2: General case (unequal variances)

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Step 3: Compute the p -value from the Test Statistic

If the null hypothesis $H_0 : \mu_1 - \mu_2 = 0$ is true, does the observed test statistic value seem “unusual” enough to reject the null hypothesis?

Definition of p -value: The p -value is the probability, assuming H_0 is true, of obtaining a test statistic $Z = 1.30$ or one more extreme.

$$Z \sim N(0, 1)$$

$$p\text{-value} = 2 \int_{-\infty}^{-1.3} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 2 \times 0.096801 = 0.1936$$

Interpretation: The p -value represents the probability of observing a difference as large (or larger) than the one seen in the data if H_0 is true.

Step 4: Compare the p -value with the Significance Level and Conclude about H_0

$$p\text{-value} = 0.19 > \alpha = 0.05 \Rightarrow \text{Fail to reject } H_0$$

Interpretation: The evidence from our data (observations and statistical testing) is not strong enough to reject the null hypothesis. Therefore, we do not have sufficient evidence to conclude that there is a significant difference in the mean BMI between males and females aged 18–29 currently living in Vietnam.

Alternative approach: Using Confidence Interval

At 5% significance level, the 95% confidence interval for $\mu_1 - \mu_2$ is:

$$(-0.37, 1.85)$$

Since 0 is within this interval, we again **fail to reject the null hypothesis**—there is no statistically significant difference between the two population means.

Lab 7

Test your own hypothesis